

Harun Ardiansyah, S.T., M.S., PhD.
Lecturer
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Republic of Indonesia Defense University
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EDUCATION

August 2021 -
December 2025

Ph.D. in Nuclear, Plasma, and Radiological Engineering
University of Illinois Urbana-Champaign, Urbana, IL, USA
PhD Dissertation Title: **A Reactor Physics Framework to Detect Anomalies in HTGR Core**
PhD. Supervisor: **Prof. Tomasz Kozlowski**

September 2019 -
May 2021

M.S. in Nuclear Engineering and Radiological Sciences
University of Michigan, Ann Arbor, MI, USA
Research Topic: **Analysis of Steady State and Transient of PBMR-400**
Advisor: **Prof. Thomas Downar**

August 2014 -
October 2018

S.T. (Equivalent to B.Eng.) in Nuclear Engineering
Universitas Gadjah Mada, Indonesia
Final Thesis Title: **Core Design Parametric Study of Integral Pressurized Water Reactor (IPWR) with Mixed Oxide Ceramic Fuel using SRAC Code System**
Final Thesis Advisor: **Dr. Alexander Agung, S.T., M.Sc.**

WORK EXPERIENCE

**April 2026 –
Present**

Lecturer
Department of Nuclear Engineering, Republic of Indonesia Defense University, Bogor, Indonesia

**August 2022 –
December 2025**

Graduate Teaching Assistant

- NPRE 455 (Neutron Diffusion and Transport),
- NPRE 247 (Modeling of Nuclear Energy Systems),
- NPRE 501 (Fundamentals of Nuclear Engineering)

**August 2022 –
December 2024**

Graduate Research Assistant
Illinois Microreactor Research, Development, and Demonstration Center (IMRD2C), University of Illinois Urbana-Champaign, Urbana, IL

**August 2021 –
December 2025**

Graduate Research Assistant
ARTS Group, University of Illinois Urbana-Champaign, Urbana, IL

- Performing analysis on neutronic and thermal hydraulic safety of micro-HTGR reactors.
- Performing application of neutron noise for advanced reactors safety.
- Developing codes for neutron noise in HTGR.

**November 2019-
May 2021**

Graduate Research Assistant
NURAM Group, University of Michigan, Ann Arbor MI

- Performed analysis on steady state and transient of PBMR-400 benchmark.

- Performed validation and verification of HTR-10 with Serpent and AGREE.
- August 2016 - May 2018** **Teaching Assistant and Laboratory Assistant**
 Universitas Gadjah Mada, Indonesia
- Radiation Detection and Measurement Lab.
 - Introduction to Nuclear Technology.

SKILLS/EXPERIENCE

- Programming:
 - Proficient: Python, MATLAB, Git.
 - Familiar: C++, MOOSE.
- Radiation Transport:
 - Monte Carlo Neutron Transport: Serpent, MCNP, OpenMC, SCALE (KENO).
 - Deterministic Neutron Transport: SRAC, PARCS.
- Thermal Hydraulics/CFD:
 - Familiar: RELAP, FLUENT
- Structural Material: ABAQUS

PROFESSIONAL ACTIVITIES

- 2025 Summer School in Open Science + Research Software Engineering by the US Research Software Sustainability Institute (URSSI), August 11-15, 2025, University of Alaska Fairbanks.
- Modeling, Experimentation, and Validation (MeV) Summer School 2024 by Argonne National Laboratory, July 29 – August 9, 2024, Argonne National Laboratory.
- Course on “Core Modeling for Transient” by GRE@T-PIONEER, virtual course, January 2024.
- Course on “Deterministic Modeling of Nuclear Reactor Multi-physics” supported by ENEN2+, December 2023, Chalmers University (Sweden).
- Nuclear Innovation Bootcamp 2022 by Nuclear Innovation Alliances, August 2022, University of Wisconsin-Madison.
- 10th International Conference on High Temperature Reactor Technology (HTR2021), virtually in June 2021. Presented a paper entitled “Evaluation of PBMR-400 Core Design Steady State Condition with Serpent and AGREE”
- Session Chair, 6th International Energy Conference ASTECHNOVA, 2021
- Indonesian Nuclear Society (HIMNI), 2021 – Present
- American Nuclear Society, 2019 - Present

HONORS AND AWARDS

- Mavis Future Faculty Fellowship, 2024 – 2025
- Best Team Project Award, MeV School, 2024
- Barclays G. Jones Graduate Fellowship, Grainger College of Engineering UIUC, 2023 – 2024
- Indonesia Endowment Fund for Education (LPDP) Scholarship, Indonesian Ministry of Finance, 2018

PUBLICATIONS

Peer-Reviewed Journals/Conference Proceedings:

1. **Ardiansyah, H., & Kozłowski, T.** (2026). Novel Neutron Noise Unfolding Methods and Applications for Anomaly Detection in 2D HTTR Core. *Proceeding of PHYSOR 2026*.

- The International Conference on Physics of Reactors (PHYSOR) 2026. <https://doi.org/10.5281/ZENODO.20803920>
2. **Ardiansyah, H.**, Kozlowski, T., Novel Neutron Noise Unfolding Methods for Anomaly Detection (To be presented in PHYSOR 2026)
 3. **Ardiansyah, H.**, & Kozlowski, T. (2026). Review of Detecting Anomalies in Power Reactors Using Neutron Noise Analysis. *Nuclear Technology*, 1–22. <https://doi.org/10.1080/00295450.2026.2619208>
 4. Dwijayanto, R.A.P., **Ardiansyah, H.**, Harto, A.W. (2024). Verification and Geometry Optimization of a One Fluid Molten Salt Reactor (OFMSR) with Fixed Volume. *ASME J of Nuclear Rad Sci.* 1-20. <https://doi.org/10.1115/1.4064465>
 5. **Ardiansyah, H.**, Oktavian, M. R. (2021). Evaluating the Diffusion Approximation Capability on the Integral Pressurized Water Reactor (IPWR) Core Calculation. *Atom Indonesia*, 47(2), 85. <https://doi.org/10.17146/aij.2021.1013>
 6. **Ardiansyah, H.**, Seker, V., Downar, T., Skutnik, S., Wieselquist, W. (2021). Evaluation of PBMR-400 Core Design Steady State Condition with Serpent and AGREE. *Journal of Physics: Conference Series*, 2048(1), 012024. <https://doi.org/10.1088/1742-6596/2048/1/012024>

Book:

1. **Ardiansyah, H.**, Eka Dewi, P., Silalahi, D. F., Gunawan, D., Wahyuni, E., Dipayana, G. F., Hardhi, M., Winofa, N. C., Ramadhan, R. A., Hidayat, T. (2022). Indonesia Post-Pandemic Outlook: Strategy towards Net-Zero Emissions by 2060 from the Renewables and Carbon-Neutral Energy Perspectives. In Penerbit BRIN. <https://doi.org/10.55981/brin.562>